

Problem Set 2 — Linear Maps

Problem 1

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the projection onto the plane $x + y + z = 0$. Write the matrix of T in the standard basis and verify $T^2 = T$.

Problem 2

Determine whether the following maps are linear, and for each linear map compute its kernel and image: (a) $f(x, y) = (x + 2y, 0)$; (b) $g(x, y) = (xy, x - y)$.

Problem 3

Suppose A is invertible and v is an eigenvector of A with eigenvalue λ . Show that v is an eigenvector of A^{-1} and find its eigenvalue.

$$A^{-1} v = \lambda^{-1} v$$

Problem 4 (bonus)

Find a 2×2 matrix that has no real eigenvalues, and explain how this is consistent with the fundamental theorem of algebra.